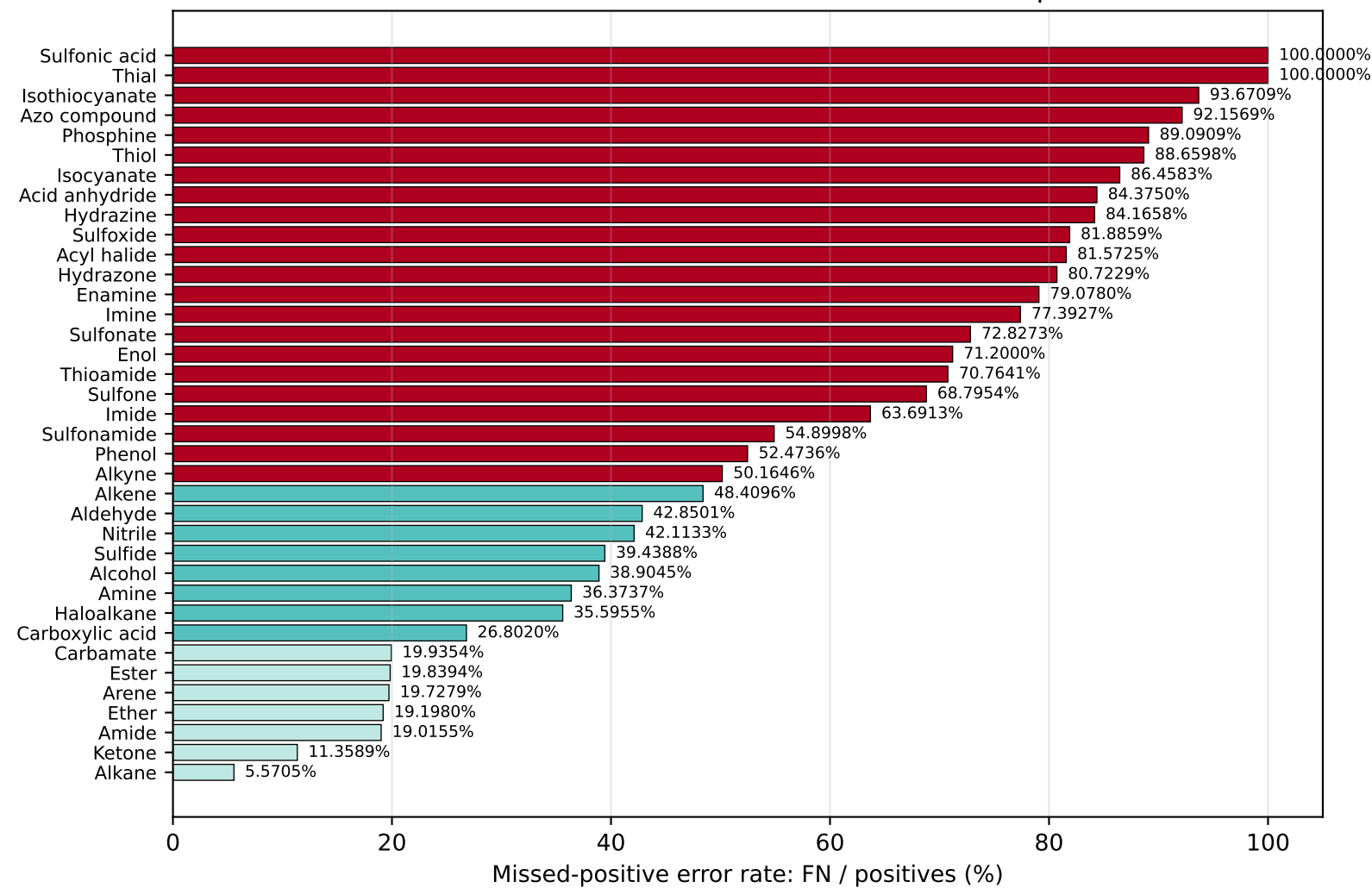
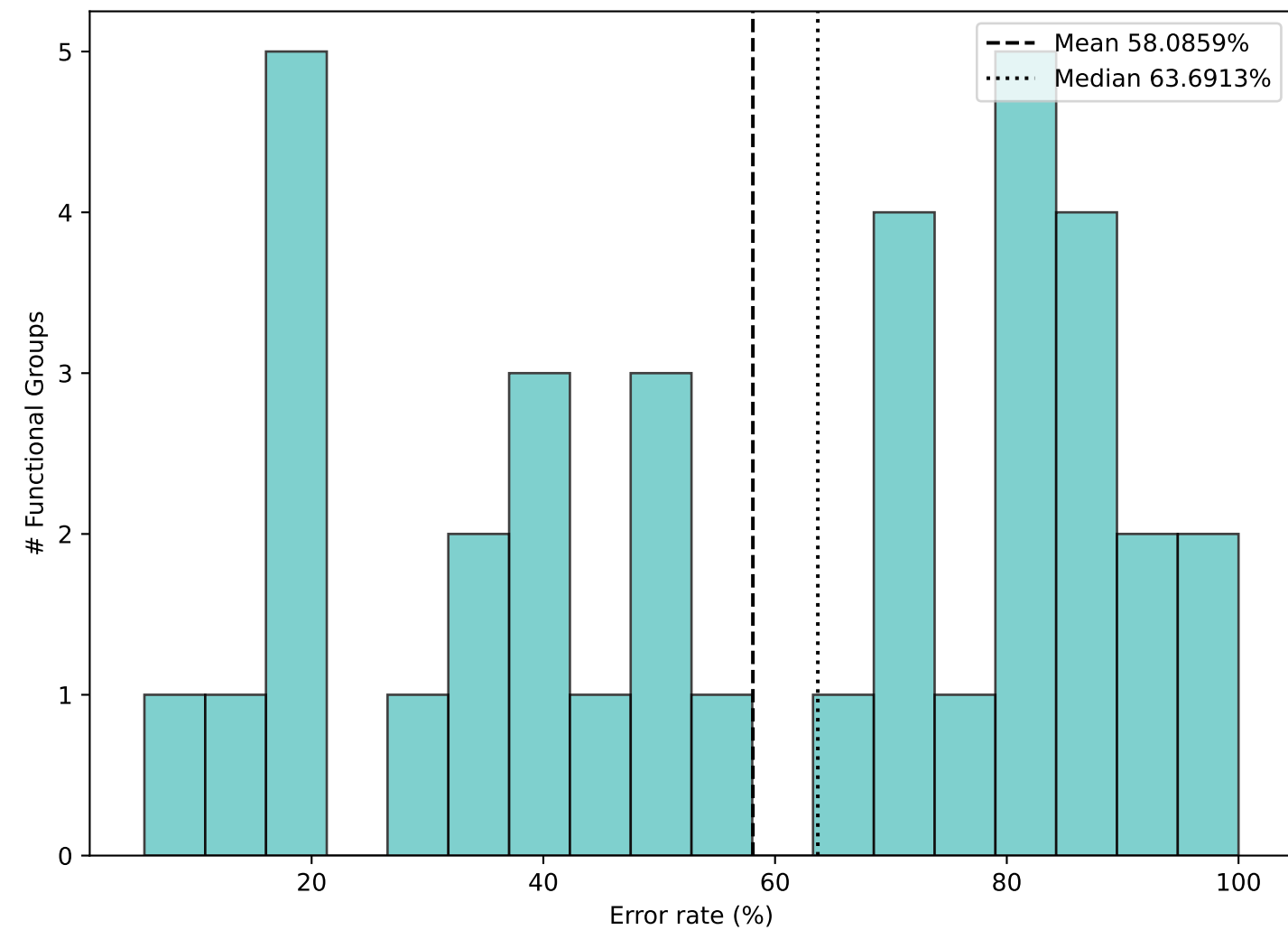


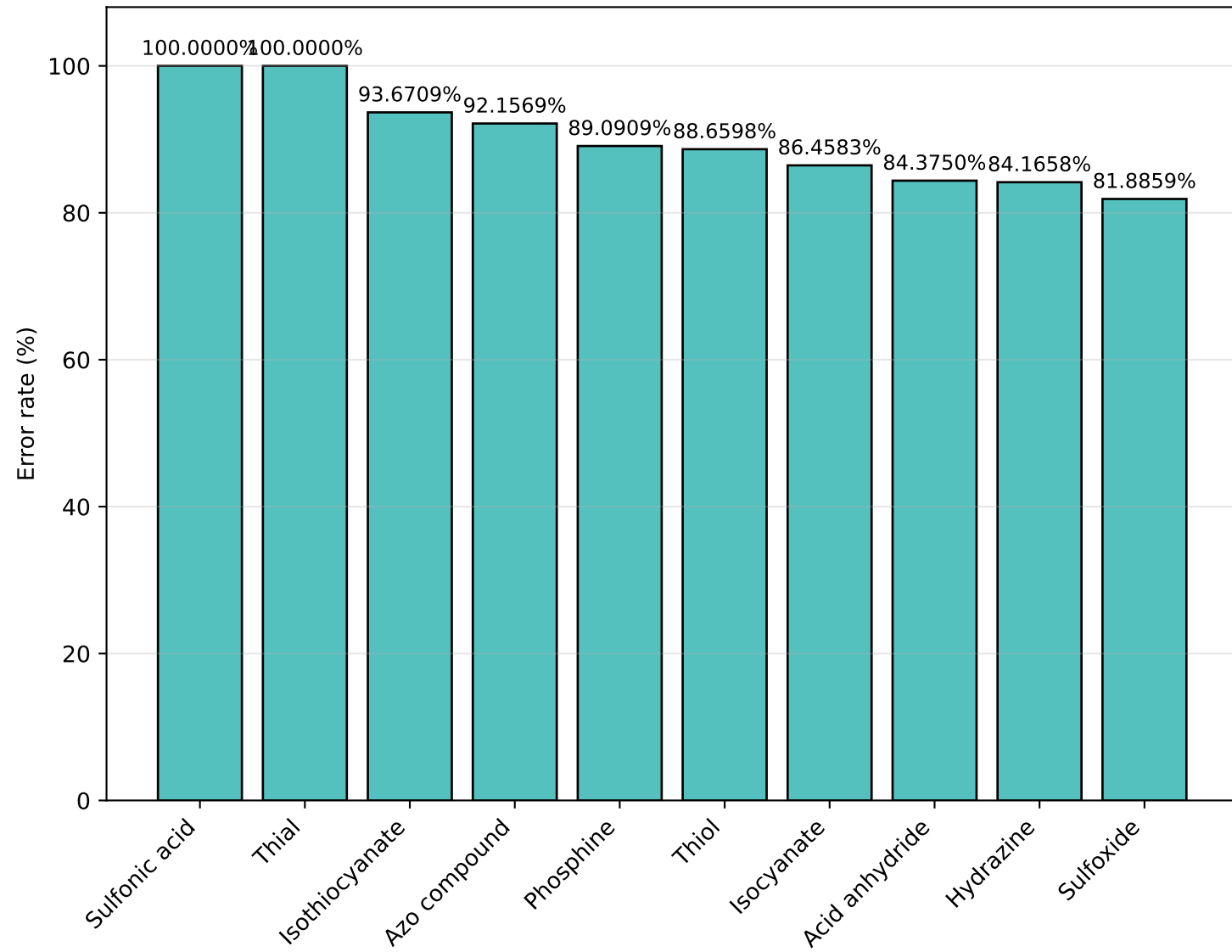
MPS C-NMR Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS C-NMR TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.6477
F1 macro: 0.3306
Precision micro: 0.5691
Recall micro: 0.7515
Hamming accuracy: 0.8943

Error-rate statistics:

Mean: 58.0859%
Median: 63.6913%
Std: 28.3233%
Max: 100.0000% (Sulfonic acid)
Min: 5.5705% (Alkane)

Problem classes:

Severe (>50%): 22 groups
High (20-50%): 8 groups
Moderate (5-20%): 7 groups
Low (<=5%): 0 groups